Data Scientist & Economic Analyst

Rodrigo França Chaves

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Website · Google Scholar · LinkedIn · GitHub · Brazilian Citizen — F1 Visa (STEM OPT Eligible)

Applied economist working on property taxation, housing markets, and utility regulation. First author of a peer-reviewed article in *Utilities Policy* estimating the health effects of water privatization across two decades of Brazilian municipal data. Builds the evidence base for regulatory and valuation questions from large administrative sources — assessment rolls, transaction records, disease registries, satellite imagery — pairing quasi-experimental identification with machine learning, and defending the result to audiences that did not build the model. Seeking full-time roles in economic consulting and applied economic analysis.

EDUCATION

M.S. in Computational Analysis and Public Policy *Sep 2025 – Jun 2027*

The University of Chicago, Chicago, IL

Specializations: Markets & Regulation; Health Policy

Coursework: Machine Learning for Public Policy, Applied Econometrics, Data Engineering, Program Evaluation, Public Finance, Statistical Computing

Postgraduate Degree in Data Science (Pós-Graduação) *Sep 2022 – Apr 2024*

Insper – Institute of Research and Education, São Paulo, Brazil

Coursework: Statistical Inference, Econometrics, Machine Learning, Cloud Computing, Predictive Modeling

B.A. in International Relations *Feb 2018 – Jun 2022*

Ibmec – Brazilian Institute of Capital Markets, Belo Horizonte, Brazil

Coursework: Political Economy, International Trade, Macroeconomics, Quantitative Methods, Public Policy, Political Science

PUBLICATIONS & WORKING PAPERS

Private Ownership of Water and Wastewater Systems: Assessing Health Impacts

Utilities Policy — Peer-Reviewed Journal Article (first author)

Evaluates whether Brazil's 2020 opening of water and sanitation systems to private operators was sound policy, comparing hundreds of closely matched municipalities before and after transfer (1998–2021) via panel econometrics and difference-in-differences. Finds private operation reduced waterborne disease on average, with the largest effects on diarrheal disease among children — direct evidence for concession design and rate-setting debates.

Resistance to Doing Right: Rising Cost of Misconduct from Body-Worn Camera Adoption

Working Paper — In Progress (co-author)

Estimates how body-worn cameras changed police behavior in São Paulo, using confidential police administrative microdata obtained for the study and a staggered rollout for identification. Preliminary results show fewer police confrontations alongside more crime recorded in heavily equipped areas, a pattern consistent with both de-policing and improved reporting; identification is still being tightened.

WORK EXPERIENCE

Data Science & Research Intern — DECDI (Formerly DIME) *Summer 2026*

World Bank, Washington, DC

- Reconstructed four decades of Mumbai property tax and land-use policy into a single consistent timeline, then tested assessment records for bunching at policy thresholds to establish whether declared values showed statistical signatures of manipulation.
- Quantified how Mumbai's market for additional floor area splits across regulatory instruments (TDR, Premium FSI, Fungible FSI), identifying which mechanism developers actually use and how market, regulatory, and political conditions drive that choice.
- Measured the urbanization response to Dakar's new Bus Rapid Transit corridor, deriving built-environment indicators from satellite imagery via remote sensing where no reliable ground-level administrative data existed. *(Findings confidential to the World Bank.)*

Graduate Research Assistant *Mar 2026 – Present*

Mansueto Institute for Urban Innovation, UChicago, Chicago, IL

- Built an end-to-end digitization pipeline — page segmentation, OCR, structured extraction — for century-old cloth-bound fire insurance registers and Library of Congress holdings, converting records that had never been machine-readable into analysis-ready data.
- Extracted thousands of firm addresses from hundreds of register pages and linked them to three historical censuses by

street, then applied NLP to firm-name patterns to classify establishments across hundreds of industries — producing a dataset of industrial location and organization in 20th-century Chicago that did not previously exist.
- Returning Sep 2026 to write the method up as a research paper documenting the pipeline for replication by other archival-data researchers.

Research Assistant *Aug 2023 – Jul 2025*

Insper Cities Lab (Arq.Futuro), São Paulo, Brazil

- Two years of quasi-experimental policy evaluation across utilities, crime, property taxation, regulation, and real estate markets, working with municipal panels running to millions of records on home prices, tax assessments, and disease incidence.
- First-authored the resulting *Utilities Policy* article on water privatization and public health, and co-authored an in-progress working paper on police body cameras.
- Reconciled administrative microdata from multiple government sources, resolving inconsistent identifiers and coverage gaps to make municipal panels comparable across two decades — the step that made the privatization analysis identifiable at all.
- Mentored undergraduate researchers on research design, econometric methodology, and academic writing; coordinated cross-functional teams under firm publication deadlines.

Teaching Assistant — GIS, Econometrics & R *Aug 2023 – Jul 2025*

Insper, São Paulo, Brazil

- Taught GIS and R to 50+ students, applying spatial analysis to São Paulo's real estate market, transit infrastructure, population density, and urban models of settlement — density and land-value gradients from the center outward.
- Advised undergraduate theses through Brazil's required capstone; one placed second in its cohort.
- Supervised an intensive field course requiring direct coordination with municipal officials.

PROJECTS

NYC Property Tax: A Horizontal-Equity Audit `Python · LightGBM · scikit-learn`

Horizontal-equity audit of New York's assessment roll, testing whether comparable properties carry comparable tax burdens. Benchmarking all 1.1M properties against their peer-group median assessed value per square foot found roughly 43% sitting more than 15% away from their peers — 245k above, 227k below. Within one peer group of 1940s Queens two-family homes worth about $750k, assessments ranged from $12.09 to $45.88 per square foot: a near-fourfold gap in effective tax rate produced by purchase date under NYC's Class 1 assessment cap rather than by any difference in the properties. Tightening peer groups from quintile to ventile value bins cut within-group price spread from $475k to $48k, trading a small F1 loss for labels that carry meaning — the judgment call the project turned on. Six peer dimensions with a five-level fallback, 138 leak-free engineered features, LightGBM, and a Streamlit prototype for audit triage. Three-person course project. [GitHub] [Live Dashboard]

Data Centers Next Door: Cloud Infrastructure and Housing Costs `R · Shiny · Spatial Econometrics`

An exploratory tool for examining how data center proximity relates to Chicago housing prices, joining georeferenced infrastructure with property transaction records in an interactive dashboard. Descriptive by design: it surfaces and maps the association for users to interrogate themselves rather than asserting a causal effect. Course project. [GitHub] [Live Dashboard]

TECHNICAL SKILLS

Methods: Causal Inference (DiD, Staggered DiD, RDD, IV), Panel & Spatial Econometrics, Program Evaluation, Predictive Modeling, Gradient Boosting, Remote Sensing

Programming & Data: Python, R, SQL, PostgreSQL, Stata; large-scale data pipelines, OCR & archival extraction, geospatial workflows

Tools: QGIS, ArcGIS, Git, LaTeX, Excel, Streamlit, Shiny

Domain: Property tax & valuation, zoning & land use, housing & infrastructure policy, utility regulation, regulatory & compliance analysis

Languages: Portuguese (Native), English (Fluent — TOEFL 112/120), Spanish (Conversational), German (Basic)